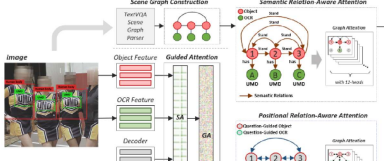


Combining TextOCR <> Semantic Relation Maps

Tuesday, July 7, 2026 9:05 PM

SceneGATE: Scene-Graph Based Co-Attention Networks for Text Visual Question Answering (2023)



Co-recognizes object and text feeding both into the graph construction

Question answering not localization

TextPlace: Visual Place Recognition and Topological Localization Through Reading

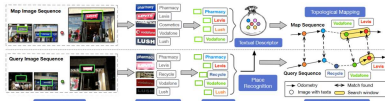
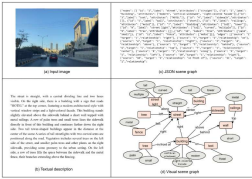


Figure 2: System overview (Top: mapping stage; Bottom: place recognition and localization stage). Scene texts are detected and recognized on both map and query image sequences. Some of the text strings are filtered by matching with the words in a pre-defined dictionary. At the mapping stage, a textual description augmented topological map is produced. It is then used for text-based place recognition and topological localization.

VPR using only text signs

Text2Graph VPR: A Text-to-Graph Expert System for Explainable Place Recognition in Changing Environments (Dec 2025)



VLLM generated description of an image -> json -> relation graph VPR (no text OCR)

GOTPR: General Outdoor Text-based Place Recognition Using Scene Graph Retrieval with OpenStreetMap (May 2025)

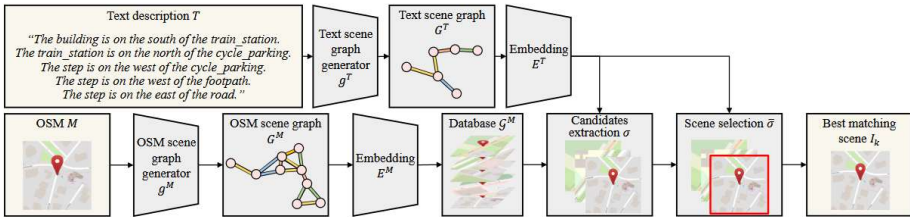


Fig. 1. Process of GOTPR. The method consists of three sequential steps: 1) Scene graph generation, 2) Scene graph candidates extraction, 3) Scene graph retrieval and scene selection. The input data are a query text description and OSM. The output is the matching scene id.

Graph matching generated from both text and osm

Bridging Text Spotting and SLAM with Junction Features (2015)

Text as SLAM landmarks.

Potential Gaps to Focus on

What method of semantic + text ocr combining has the best localization performance?

Post-hoc augmentation of detector-generated semantic relation graphs with independently recognized scene text, followed by text-aware graph matching for coarse localization

Pure semantic graph + ocr localization is somehow unique?

Research has repeatedly combined object semantics, scene text and localization, but mostly through metric semantic maps, independent landmark channels, descriptor fusion or generated language. Explicitly grounded object-OCR relation graphs used as the sole persistent map for graph-to-graph place recognition appear much less established.

Potential Approaches

Utilize bounding box overlap if that can be extracted

Refinement:

If DEMTR -> EGTR is too computational complex use a lower resource non transformation/attention model. Take out the vision backbone and add a relation generator on top. This could even be hierarchical for live SLAM.

Limitations:

Localization from a highly zoomed in view may not work as detection of a single object such as a door is not meaningfully distinct information. Detection of multiple objects is necessary.

ADD A NOTIFICATION THAT INSUFFICIENT INFORMATION IS IN FRAME. Or that additional context is needed, "Swing the phone around."